Ex. NO : 05 Date : 6/8/2026

Experiments on packet capture tool: Wireshark.

Aim :

Experiments on packet capture tool : wireshark.

Packet sniffer :

*. sniffs messages being sent / received from/by your computer.

*. store & display the contents of the various protocol fields in the messages.

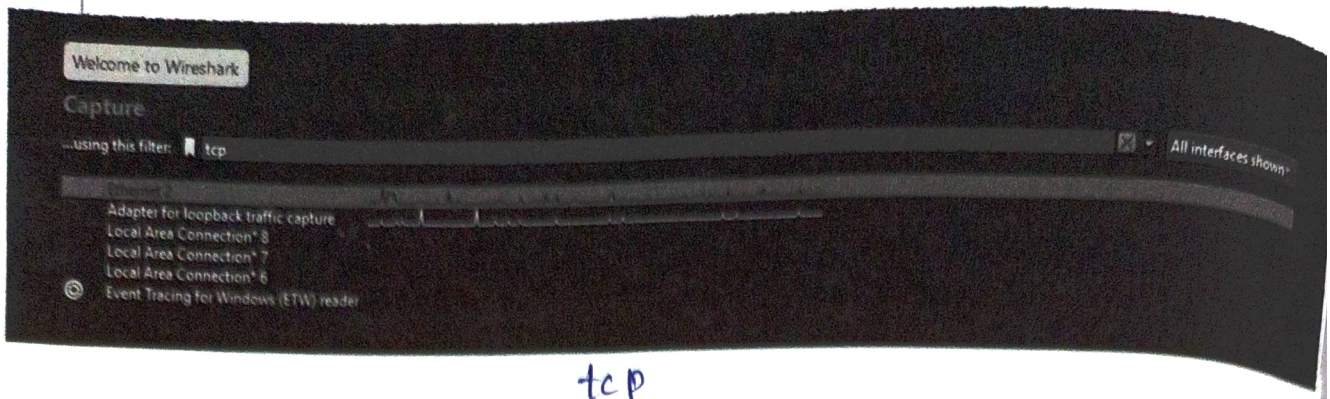
*. Passive program

- never sends packets itsels
- no packets addressed to it
- receives a copy of all packets (sent / received).

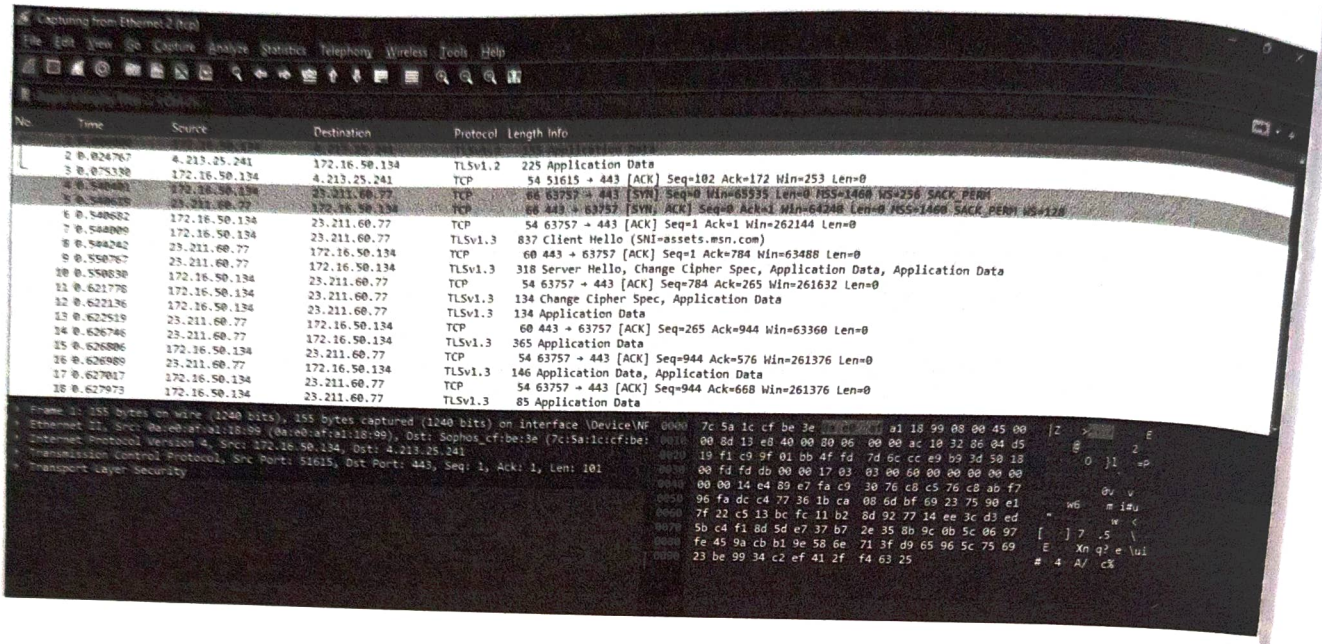
Theory :

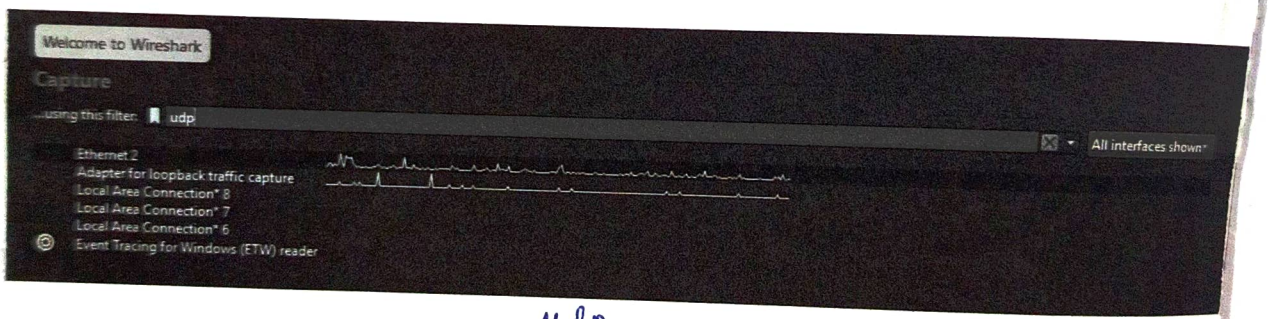
*. Wire shark is a network packet analyses used to capture and input network traffic in real time.

*. It helps in troubleshooting network issues analyzing protocols, monitoring traffic and learning network communication.

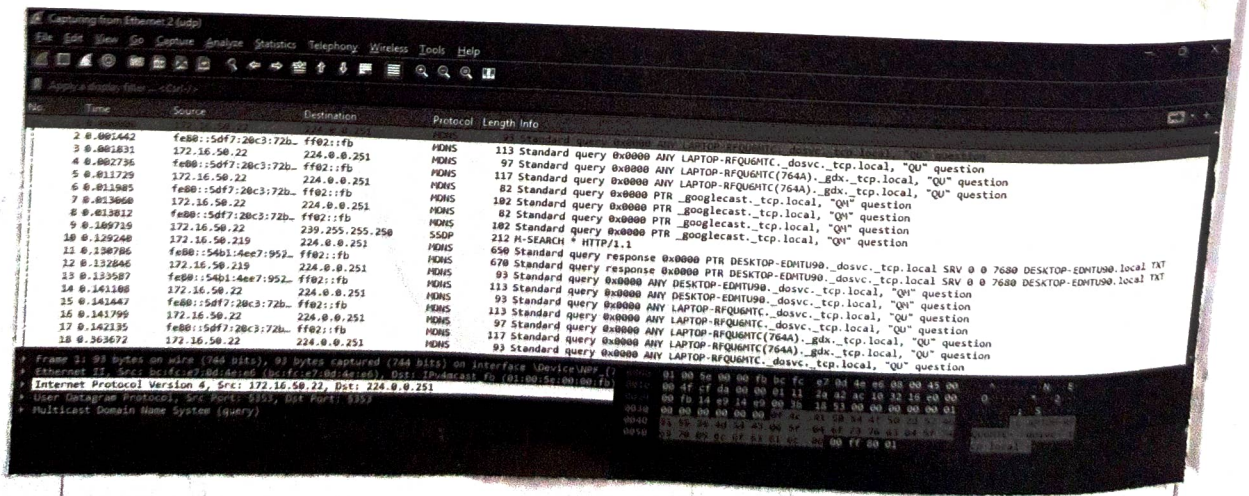

Welcome to Wireshark
Capture
...using this filter: tcp
All interfaces shown
Adapter for loopback traffic capture
Local Area Connection* 8
Local Area Connection* 7
Local Area Connection* 6
Event Tracing for Windows (ETW) reader

tcp


Capturing from Ethernet 2 (tcp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
2 0.024767 4.213.25.241 172.16.50.134 TLSv1.2 225 Application Data
3 0.075330 172.16.50.134 4.213.25.241 TCP 54 51615 → 443 [ACK] Seq=102 Ack=172 Win=253 Len=0
4 0.540481 172.16.50.134 23.211.60.77 TCP 66 63757 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
5 0.540619 23.211.60.77 172.16.50.134 TCP 66 443 → 63757 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
6 0.540682 172.16.50.134 23.211.60.77 TCP 54 63757 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
7 0.544009 172.16.50.134 23.211.60.77 TLSv1.3 837 Client Hello (SNI=assets.msn.com)
8 0.594242 23.211.60.77 172.16.50.134 TCP 60 443 → 63757 [ACK] Seq=1 Ack=784 Win=63488 Len=0
9 0.550767 23.211.60.77 172.16.50.134 TLSv1.3 318 Server Hello, Change Cipher Spec, Application Data, Application Data
10 0.550830 172.16.50.134 23.211.60.77 TCP 54 63757 → 443 [ACK] Seq=784 Ack=265 Win=261632 Len=0
11 0.621778 172.16.50.134 23.211.60.77 TLSv1.3 134 Change Cipher Spec, Application Data
12 0.622136 172.16.50.134 23.211.60.77 TLSv1.3 134 Application Data
13 0.622519 23.211.60.77 172.16.50.134 TCP 60 443 → 63757 [ACK] Seq=265 Ack=944 Win=63360 Len=0
14 0.626746 23.211.60.77 172.16.50.134 TLSv1.3 365 Application Data
15 0.626806 172.16.50.134 23.211.60.77 TCP 54 63757 → 443 [ACK] Seq=944 Ack=576 Win=261376 Len=0
16 0.626989 23.211.60.77 172.16.50.134 TLSv1.3 146 Application Data, Application Data
17 0.627017 172.16.50.134 23.211.60.77 TCP 54 63757 → 443 [ACK] Seq=944 Ack=668 Win=261376 Len=0
18 0.627973 172.16.50.134 23.211.60.77 TLSv1.3 85 Application Data
Frame 1: 155 bytes on wire (1240 bits), 155 bytes captured (1240 bits) on interface \Device\NF
Ethernet II, Src: 0a:e0:af:a1:18:99 (0a:e0:af:a1:18:99), Dst: Sophos_cf:be:3e (7c:5a:1c:cf:be:
Internet Protocol Version 4, Src: 172.16.50.134, Dst: 4.213.25.241
Transmission Control Protocol, Src Port: 51615, Dst Port: 443, Seq: 1, Ack: 1, Len: 101
Transport Layer Security


Welcome to Wireshark
Capture
...using this filter: udp
All interfaces shown
Ethernet 2
Adapter for loopback traffic capture
Local Area Connection* 8
Local Area Connection* 7
Local Area Connection* 6
Event Tracing for Windows (ETW) reader

udp


Capturing from Ethernet 2 (udp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
2 0.001442 fe80::5df7:20c3:72b… ff02::fb MDNS 113 Standard query 0x0000 ANY LAPTOP-RFQU6MTC._dosvc._tcp.local, "QU" question
3 0.001831 172.16.50.22 224.0.0.251 MDNS 97 Standard query 0x0000 ANY LAPTOP-RFQU6MTC(764A)._gdx._tcp.local, "QU" question
4 0.002736 fe80::5df7:20c3:72b… ff02::fb MDNS 117 Standard query 0x0000 ANY LAPTOP-RFQU6MTC(764A)._gdx._tcp.local, "QU" question
5 0.011729 172.16.50.22 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
6 0.011985 fe80::5df7:20c3:72b… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
7 0.013060 172.16.50.22 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
8 0.013812 fe80::5df7:20c3:72b… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
9 0.109719 172.16.50.22 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
10 0.129248 172.16.50.219 224.0.0.251 MDNS 650 Standard query response 0x0000 PTR DESKTOP-EDMTU90._dosvc._tcp.local SRV 0 0 7680 DESKTOP-EDMTU90.local TXT
11 0.130786 fe80::54b1:4ee7:952… ff02::fb MDNS 670 Standard query response 0x0000 PTR DESKTOP-EDMTU90._dosvc._tcp.local SRV 0 0 7680 DESKTOP-EDMTU90.local TXT
12 0.132846 172.16.50.219 224.0.0.251 MDNS 93 Standard query 0x0000 ANY DESKTOP-EDMTU90._dosvc._tcp.local, "QM" question
13 0.133587 fe80::54b1:4ee7:952… ff02::fb MDNS 113 Standard query 0x0000 ANY LAPTOP-RFQU6MTC._dosvc._tcp.local, "QM" question
14 0.141108 172.16.50.22 224.0.0.251 MDNS 93 Standard query 0x0000 ANY DESKTOP-EDMTU90._dosvc._tcp.local, "QM" question
15 0.141447 fe80::5df7:20c3:72b… ff02::fb MDNS 113 Standard query 0x0000 ANY LAPTOP-RFQU6MTC._dosvc._tcp.local, "QU" question
16 0.141799 172.16.50.22 224.0.0.251 MDNS 97 Standard query 0x0000 ANY LAPTOP-RFQU6MTC(764A)._gdx._tcp.local, "QU" question
17 0.142135 fe80::5df7:20c3:72b… ff02::fb MDNS 117 Standard query 0x0000 ANY LAPTOP-RFQU6MTC(764A)._gdx._tcp.local, "QU" question
18 0.363672 172.16.50.22 224.0.0.251 MDNS 93 Standard query 0x0000 ANY LAPTOP-RFQU6MTC._dosvc._tcp.local, "QU" question
Frame 1: 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface \Device\NPF_{
Ethernet II, Src: bc:fc:e7:0d:4e:e6 (bc:fc:e7:0d:4e:e6), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
Internet Protocol Version 4, Src: 172.16.50.22, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (query)

*. It captures packets, decodes protocols, applies filters display, statistics and analyze packet flow.

Features :

*. capture live network traffic.
*. support protocols decoding.
*. provides captures & display features
*. Displays packet details & statistics.
*. Generates flow graphs.

Wireshark Interface :

1) Packet list - display all captured Packet.

2) Packet details - shows the protocol information of the selected packet

3) Packet bytes - Display packet data in hexa - decimal format.

Common Filter :

*. tcp - displays tcp packets
*. Udp - displays Udp packets
*. arp - displays arp packets
*. dns - displays dns packets
* http - displays http packets.
*. icmp - displays icmp packets
*. dhcp - displays dhcp packets

Procedure :

*. Open wireshark

*. Select the ethernet / local area Connection Interface.

*. Go to capture → options

*. set stop capture after 100 packets

* click start to begin packet capture

*. Apply the required filter (Tcp, Udp, Arp, DNS, Http...).

*. Inspect the captured packet.

*. For Tcp, DNS, view statistics -> flow graph.

*. Save the captured packets.

Tcp and udp packets :

*. Applied the tcp and udp display filter in wireshark to view only TCP & UDP packets.

*. Observed communication between the source and destination using IP address & Port number.

*. Examined packet detail such as source / destination, address, ports, protocal address and payload.

*. viewed the communication sequence using statistics -> flow graph.

* Successfully analyzed and saved the captures.

Result:

Successfully captured, filtered and analyzed different network packets using wireshark. The tcp and udp packets helped understand the communication between network devices and network protocols.